Hannah Friedman

Email: hannahfriedman@berkeley.edu
Website: hannahyfriedman.com

ICERM
Brown University, Providence, RI

EMPLOYMENT

Georgia Institute of Technology August 2027 – May 2030
Hale Visiting Assistant Professor

Max Planck Institute for Mathematics in the Sciecnes May 2027 – July 2027
Postdoc

Institute for Computational and Experimental Research in Mathematics September 2026 – May 2027
Institute Postdoctoral Fellow

EDUCATION

University of California, Berkeley May 2026
PhD in Mathematics
Jointly advised by Bernd Sturmfels and Serkan Hoşten
Thesis: *Metric Algebraic Geometry of Grassmannians*

Harvey Mudd College May 2023
BS in Mathematics with High Distinction, Honors in Mathematics
Advised by Michael Orrison
Thesis: *Long Increasing Subsequences*

AWARDS & HONORS

Chancellor's Fellowship, UC Berkeley 2023

Alvin White Prize, Harvey Mudd College 2023
Presented to a student or students who have contributed greatly to the humanistic side of the Harvey Mudd College Mathematics Community.

Giovanni Borrelli Mathematics Fellowship, Harvey Mudd College 2021 - 2022
Conferred on one or two junior mathematics majors in recognition of the ability to complete high quality research independently or in tandem with a faculty advisor.

Courtney S. Coleman Prize, Harvey Mudd College 2021
Awarded to three rising juniors who excel in mathematics.

RESEARCH ACTIVITES

- Visit to Max Planck Institute for Mathematics in the Sciences — Leipzig, Summers 2023-2026
- Geometric Methods in Optimization and Numerical Analysis — MFO Oberwolfach, July, 2026
- Visit to Academy of Mathematics and Systems Science, CAS — Beijing, July 2026
- Organized Metric Algebraic Geometry: Starting Local — MPI MiS, April 2026
- Combinatorical Coworkspace — Kleinwalsertal, March 2026
- SL Math Summer School: New Perspectives on Discriminants and Applications — MPI MiS, June 2025
- Problem-Solving Workshop: Computational Geometric Analysis — CUNY Graduate Center, May 2025
- Workshop on the Applications of Commutative Algebra — The Fields Institute, May 2025
- Organized Non-Linear Algebra Seminar — UC Berkeley, December 2024
- Workshop for Women in Algebraic Statistics — St John's College, Oxford, July 2024

PUBLICATIONS

13. Hannah Friedman, Serkan Hoşten, Andrea Rosana. *Distance Optimization Within the Grassmannian*. Submitted (2026).

12. Hannah Friedman. *Maximum Likelihood Estimation on the Grassmannian of Lines*. Submitted (2026).

11. Hannah Friedman, Andrea Rosana, Bernd Sturmfels. *Distance Optimization in the Grassmannian of Lines*. Submitted (2026).

10. Viktoriia Borovik, Hannah Friedman, Serkan Hoşten, Max Pfeffer. *Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties*. Submitted (2025).

9. Renato G. Bettiol, Hannah Friedman. *Counting Homogeneous Einstein Metrics*. Submitted (2025).

8. Paul Breiding, Taylor Brysiewicz, Hannah Friedman. *Homotopy Iterators*. Submitted (2025).

7. Hannah Friedman, Bernd Sturmfels, Max Wiesmann. *Squared Linear Models*. SIAM Journal on Applied Algebra and Geometry **10** (2026) 153–173.

6. Hannah Friedman, Serkan Hoşten. *Grassmann and Flag Varieties in Linear Algebra, Optimization, and Statistics: An Algebraic Perspective*. Advances in Applied Mathematics **180** (2026) 103130.

5. Yulia Alexandr, Kristen Dawson, Hannah Friedman, Fatemeh Mohammadi, Pardis Semnani, Teresa Yu. *Decomposing Conditional Independence Ideals with Hidden Variables*. Advances in Applied Mathematics **174** (2026) 103003.

4. Hannah Friedman. *Likelihood Geometry of the Squared Grassmannian*. Proceedings of the American Mathematical Society **153** (2025) 4463–4474.

3. Karel Devriendt, Hannah Friedman, Bernhard Reinke, Bernd Sturmfels. *The Two Lives of the Grassmannian*. Acta Universitatis Spientiae, Mathematica, **17** (2025) 8.

2. Hannah Friedman, Bernd Sturmfels, Max Zubkov. *Likelihood Geometry of Determinantal Point Processes*. Algebraic Statistics, **15** (2024) 15–25.

1. Hannah Friedman, Amani R. Maina-Kilaas, Julianna Schalkwyk, Hina Ahmed, and Jamie Haddock. *An Interpretable Joint Nonnegative Matrix Factorization-Based Point Cloud Distance Measure*, Proceedings of the Conference on Information Sciences and Systems (CISS), Baltimore, MD, March 2023.

TALKS

Key: ♡ = colloquium; ♣ = invited; ♠ = contributed; ♢ = seminar/lecture.

- Distance Optimization Within the Grassmannian
  ♢ Math and Data Seminar, University of Missouri–Columbia, October 2026.
- Maximum Likelihood Estimation for Statistical Models on the Grassmannian.
  ♣ Special Session on Applied Algebra in Statistics, Optimization, and Machine Learning, AMS Southeastern Sectional Meeting, October 2026.
  ♠ Geometric Methods in Optimization and Numerical Analysis, Oberwolfach, July 2026.
- Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties.
  ♠ Effective Methods in Algebraic Geometry, July 2026.
- The ML Degree of the Grassmannian
  ♠ Metric Algebraic Geometry at Institut Mittag-Leffler, June 2026.
- Metric Algebraic Geometry of the Grassmannian of Lines.
  ♢ Oberseminar at Universität Magdeburg, April 2026.
  ♢ Algebra Seminar at Universität Osnabrück, April 2026.
- Distance Optimization.
  ♢ Metric Algebraic Geometry: Starting Local, April 2026.

- Counting Homogeneous Einstein Metrics.
  ♡ Santa Clara University Mathematics and Computer Science Colloquium, November 11, 2025.
  ♣ San Francisco State University Algebra, Combinatorics, and Geometry Seminar, October 8, 2025.
  ♢ UC Berkeley Seminar on Commutative Algebra and Algebraic Geometry, September 30, 2025.
- Grassmann and Flag Varieties in Linear Algebra, Optimization, and Statistics: An Algebraic Perspective.
  ♣ New Frontiers of Geometry and Combinatorics in Machine Learning at SIAM AG, July 10, 2025.
- Squared Linear Models
  ♢ MPI MiS Seminar on Nonlinear Algebra, June 19, 2025.
- Solving Polynomial Equations (Software Demsonstration)
  ♣ Problem Solving Workshop: Computational Geometric Analysis, June 4, 2025.
- Likelihood Geometry of the Squared Grassmannian
  ♣ Session on Advances in Applied Algebra and Algebraic Statistics at the AWM Resarch Symposium, March 27, 2025.
  ♠ AlgStat 2025, March 27, 2025.
  ♣ AMS Special Session on Algebraic Statistics in Our Changing World at JMM, January 8, 2025.
  ♢ UC Berkeley Combinatorics Seminar, October, 2024.
  ♠ Workshop for Women in Algebraic Statistics at St. John's College, Oxford, July, 2024.
- Likelihood Geometry of the Squared Grassmannian
  ♢ University of Wisconsin Applied Algebra Seminar, May 15, 2025.
- The Squared Grassmannian
  ♣ University of Washington Combinatorics and Geometry Seminar, January 15, 2025.
- What is algebraic statistics?
  ♠ Bay Area Discrete Math Day, December 7, 2024.
- Optimization over Two Embeddings of the Grassmannian
  ♣ Codex seminar, July 2024.
- Likelihood Geometry of Determinantal Point Processes
  ♢ Algebra-Geometry-Combinatorics Afternoon at SFSU, November 2023.
- Long Increasing Subsequences
  ♠ Harvey Mudd Thesis Presentation Days.
- A New Basis for k-Local Class Functions
  ♠ 2023 Joint Math Meetings.

TEACHING & MENTORSHIP

- Unbounded Representation Leader, Fall 2024 – Fall 2025
- Graduate Student Instructor at UC Berkeley:
  - Math 16B: Analytic Geometry and Calculus, Fall 2025
  - Math 250B: Graduate Commutative Algebra, Spring 2025
  - Math H110: Honors Abstract Linear Algebra, Spring 2025
  - Math 55: Discrete Math, Fall 2024
  - Math 1A: Calculus I, Spring 2024
- UC Berkeley Directed Reading Program Mentor, Spring 2024 & Fall 2025
- President of Women in Math at Harvey Mudd College, Fall 2022-Spring 2023

SKILLS

- Programming Languages: Python, MatLab, Macaulay2, Java, C++, R, Julia
- Languages: English (fluent), German (fluent), Mandarin (proficient), Hebrew (proficient)